Outlier Analysis

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Outline

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- Probabilistic Models
- Clustering for Outlier Detection
- Distance-Based Outlier Detection
- Density-Based Methods
- Information-Theoretic Models
- Outlier Validity

Introduction

- An outlier is a data point that is very different from most of the remaining data.

- “*An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism.*” (Hawkins)

- Outliers are also referred to as abnormalities, discordants, deviants, or anomalies

Introduction

- Outliers can be viewed as a complementary concept to that of clusters.
  - Clustering attempts to determine groups of data points that are *similar*.
  - Outliers are individual data points that are *different* from the remaining data.

- It serves for different applications
  - Data cleaning → remove noise in data
  - Credit card fraud → unusual patterns of credit card activity
  - Network intrusion detection → unusual records/changes in network traffic

Introduction

- Most outlier detection methods create a model of normal patterns. Outliers are defined as data points that do not naturally fit within this normal model.

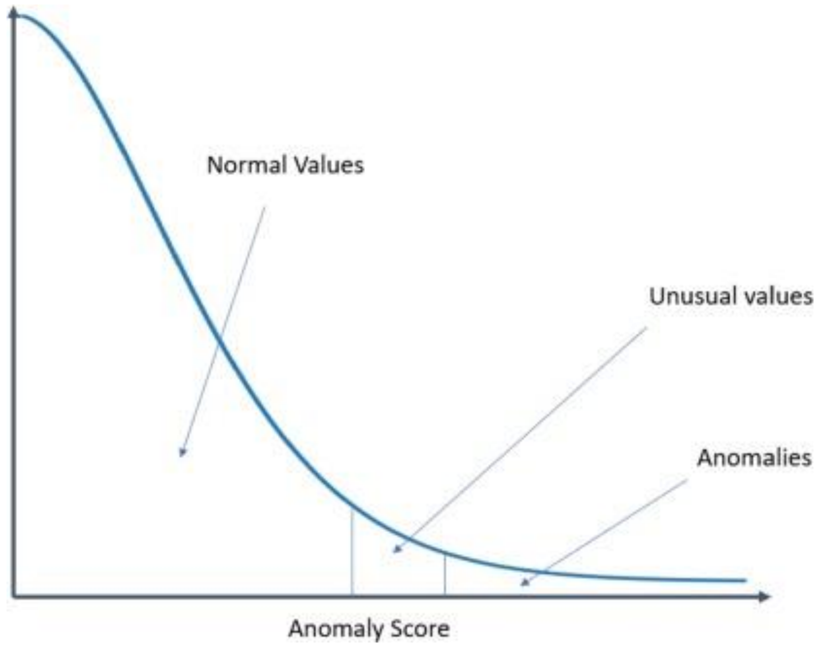
- The *outlierness* of a data point is quantified by a numeric value, the outlier score.
  - Real-valued outlier score: it quantifies the tendency for a data point to be considered an outlier. It can be a probability value quantifying the likelihood that a given data point is an outlier.
  - Binary label: it indicates whether or not a data point is an outlier. It is less informative that real-valued scores.

Introduction

- The generation of an outlier score requires the construction of a model of the normal patterns. A model may be designed to produce specialized types of outliers based on a very restrictive model of normal patterns.
- Key models for outlier analysis:
    - Probabilistic Models
    - Clustering for Outlier Detection
    - Distance-Based Outlier Detection
    - Density-Based Methods
    - Information-Theoretic Models

Extreme Value Analysis

- Extreme values are a specific kind of outliers.
- Such outliers correspond to the statistical tails of probability distributions.
  - The 1-dimensional case is the more common, but their definition can be extended to the multidimensional case.
- All extreme values are outliers, but the reverse may not be true.



Extreme Value Analysis

- Consider the 1-dimensional data set

$$\{1,3,3,3,50,97,97,97,100\}$$

- What are the extreme values? What are the outliers?

Extreme Value Analysis

- Consider the 1-dimensional data set

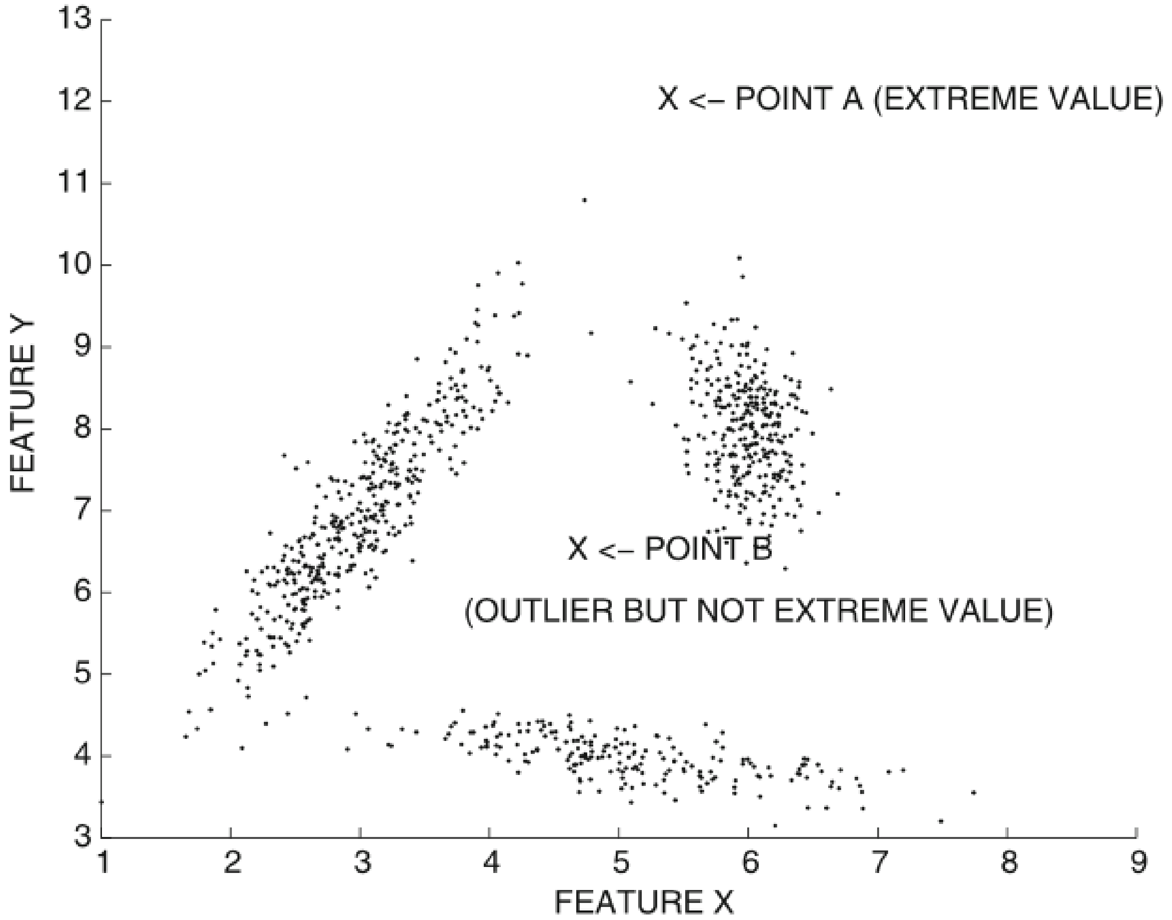
$$\{1,3,3,3,50,97,97,97,100\}$$

- The values 1 and 100 may be considered extreme values.
- The value 50 is the mean of the data set and is therefore not an extreme value.
- But, 50 is the most isolated point in the data set and should, be considered an outlier from a generative perspective.

Extreme Value Analysis

- In a multivariate setup, is more difficult to define the multivariate tail area.

- All extreme values are outlies
- Outlies may not be extreme values



Extreme Value Analysis

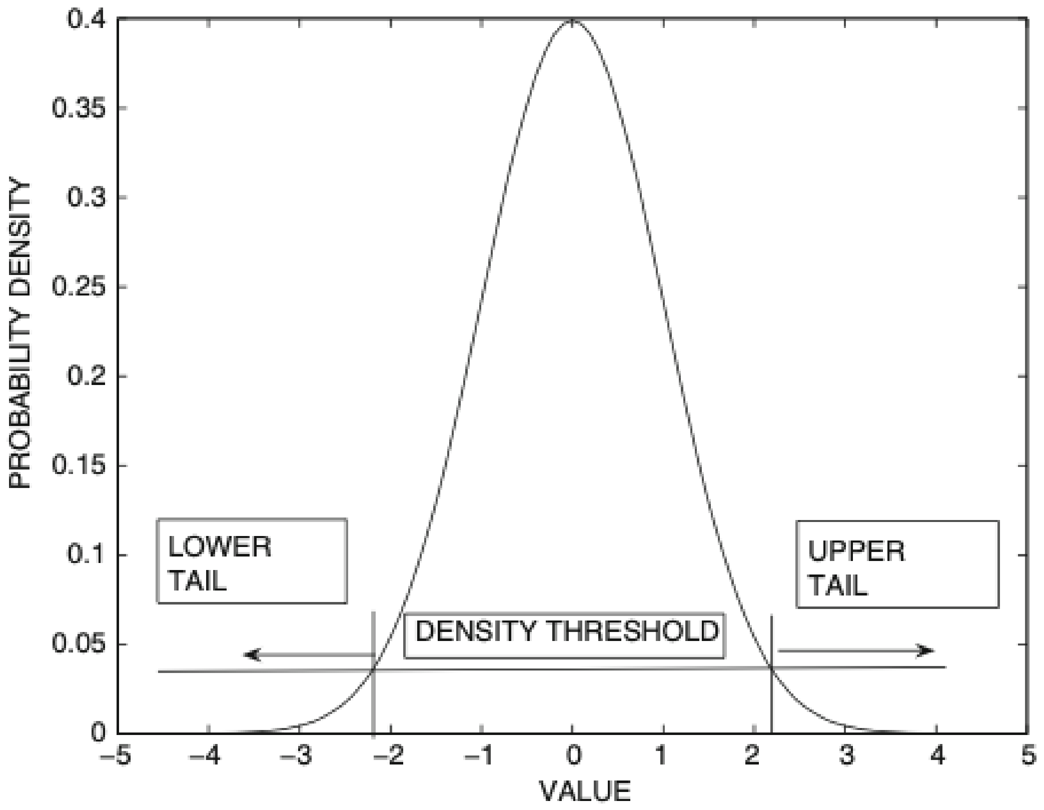
- Methods for Extreme Value Analysis
  - Univariate Extreme Value Analysis
  - Multivariate Extreme Values Analysis
  - Depth-Based Methods

Univariate Extreme Value Analysis

- Univariate extreme value analysis is intimately related to the notion of statistical tail confidence tests.
- Consider the 1-dimensional data case, in which data are described by a specific distribution function $f_X(x)$.
- Tails are extreme regions where $f_X(x) \leq \theta$

Univariate Extreme Value Analysis

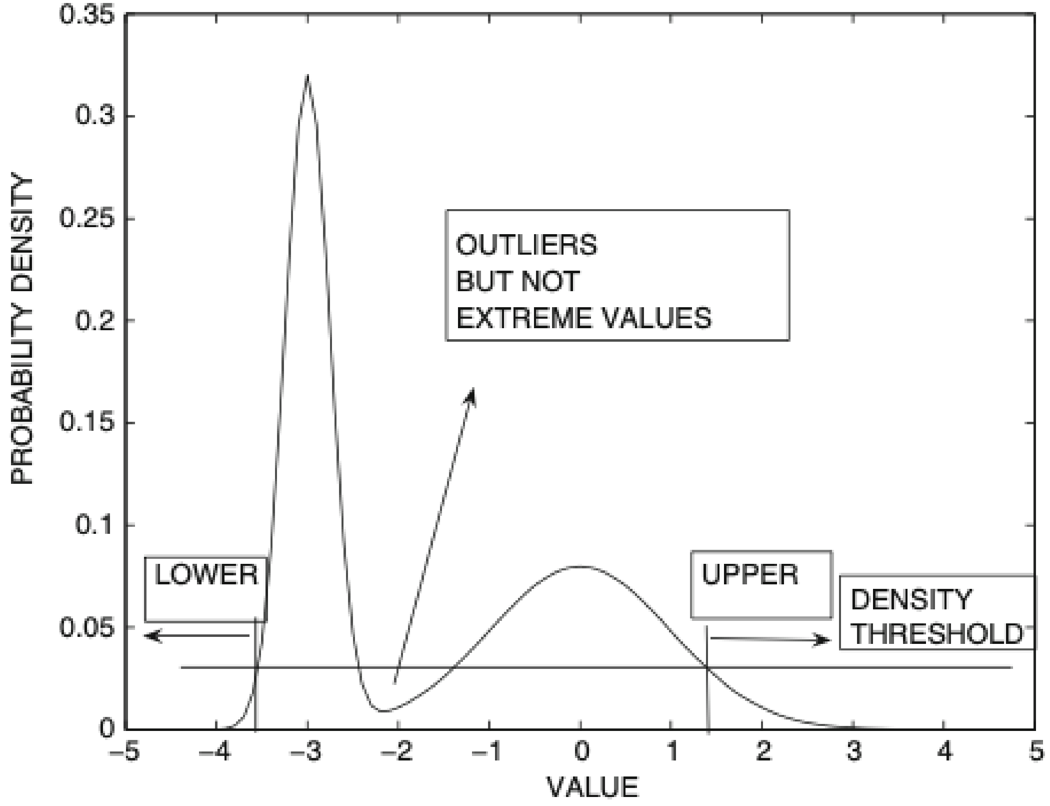
- Symmetric distribution
  - Two symmetric tails
  - Tails can be defined in terms of area rather than density threshold
  - The areas inside tails represent the cumulative probability



(a) Symmetric distribution

Univariate Extreme Value Analysis

- Asymmetric distribution
  - Areas in two tails are different
  - Regions in the interior are not tails
  - A model is needed to be defined to determine tails' probabilities



(b) Asymmetric distribution

Univariate Extreme Value Analysis

- A model distribution is selected
  - Normal Distribution with mean μ and standard deviation σ
  - Mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$ can be estimated from data or calculated via prior domain knowledge

- Z value can be calculated (normalization)

$$z_i = \frac{x_i - \mu}{\sigma}$$

  - Large positive values of z_i correspond to the upper tail
  - Large negative values of z_i correspond to the lower tail

Univariate Extreme Value Analysis

- z_i follows the standard normal distribution

$$fx(z_i) = \frac{1}{\sigma \cdot \sqrt{2 \cdot \pi}} \cdot e^{\frac{-z_i^2}{2}}$$

- Extreme values can be determined as $|z_i| \geq \tau$
- As a rule of thumb, if $|z_i| > 3$, the corresponding data points are considered extreme values.

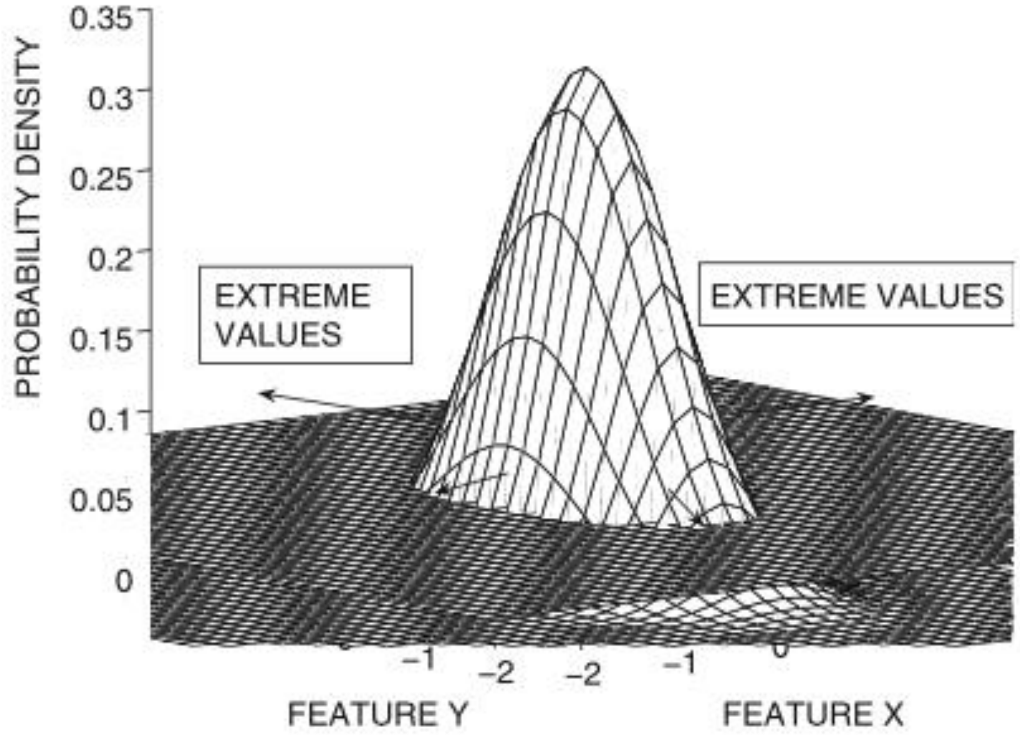
Multivariate Extreme Values Analysis

- In case of multivariate extreme values analysis, tails can be defined with a multivariate Gaussian models, which parameters can be estimated.
- Data points that are as far away as possible from the center of the cluster, in all directions, should be considered extreme values.

$$
\begin{aligned}
f(\overline{X}) &= \frac{1}{\sqrt{|\Sigma|} \cdot (2 \cdot \pi)^{(d/2)}} \cdot e^{-\frac{1}{2} \cdot (\overline{X} - \overline{\mu}) \Sigma^{-1} (\overline{X} - \overline{\mu})^T} \\
&= \frac{1}{\sqrt{|\Sigma|} \cdot (2 \cdot \pi)^{(d/2)}} \cdot e^{-\frac{1}{2} \cdot Maha(\overline{X}, \overline{\mu}, \Sigma)^2}
\end{aligned}
$$

Multivariate Extreme Values Analysis

- The obtained Mahalanobis distance from data point $\bar{X}$ to the mean of the data $\bar{\mu}$ can be used as an extreme value score of $\bar{X}$.
  - Larger values imply more extreme behavior
  - Mahalanobis distance is equal to the Euclidean distance after a PCA transformation



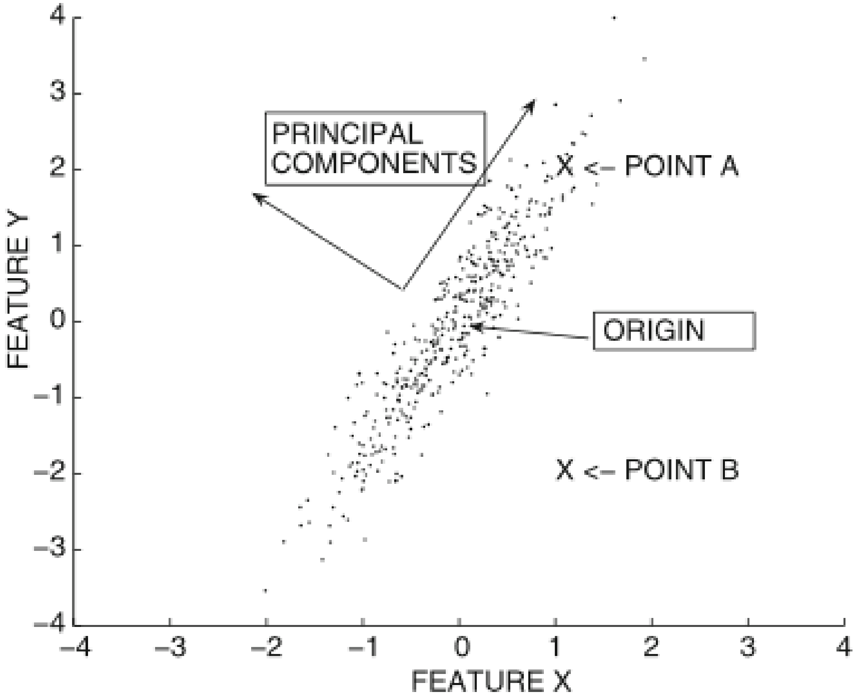
(b) Multivariate extreme values

Multivariate Extreme Values Analysis

- The extreme value probability of $\bar{X}$, is the cumulative probability of the multidimensional region for which the Mahalanobis distance to the mean $\bar{\mu}$ of the data is greater than the one between $\bar{X}$ and $\bar{\mu}$.
  - Each independent component of the Mahalanobis distances, is modeled as 1-d standard normal distribution with $\mu = 0$ and $\sigma = 1$.
  - The sum of the squares of *d* variables, drawn, will result in a variable drawn from an χ^2distribution with *d* degrees of freedom.
  - The cumulative probability of the region, for which the value is greater than $Maha(\bar{X}, \bar{\mu}, \Sigma)$, is the extreme value probability of $\bar{X}$.
- Smaller values of the probability imply greater likelihood of being an extreme value.

Multivariate Extreme Values Analysis

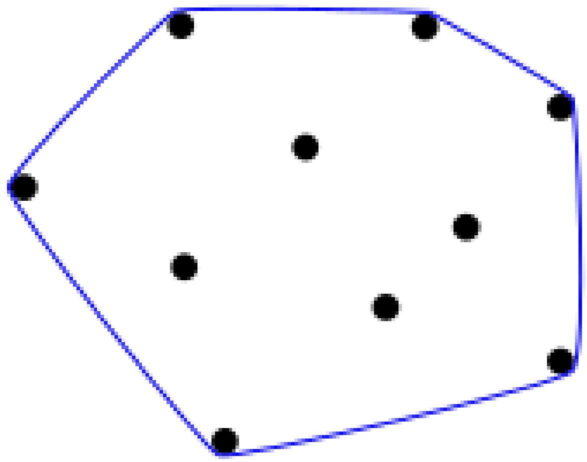
- Data distribution is modeled along uncorrelated directions
  - statistically independent normal and standardized distributions
- Data point B can be more considered a multivariate extreme value than data point A, as for natural correlations of data.
- B is closer to the centroid of the data than A, as for Euclidean distance but not for the Mahalanobis distance.



Depth-Based Methods

- Depth-based methods are based on the general principle that the convex hull of a set of data points represents the pareto-optimal extremes of this set.
- A convex hull of a set C is the set of all convex combinations of points in C

$$\operatorname{conv} C = \{\theta_1 x_1 + \cdots + \theta_k x_k |$$
$$x_i \in C, \theta_i \geq 0,$$
$$i = 1, \ldots, k,$$
$$\theta_1 + \ldots + \theta_k = 1\}$$



Depth-Based Methods

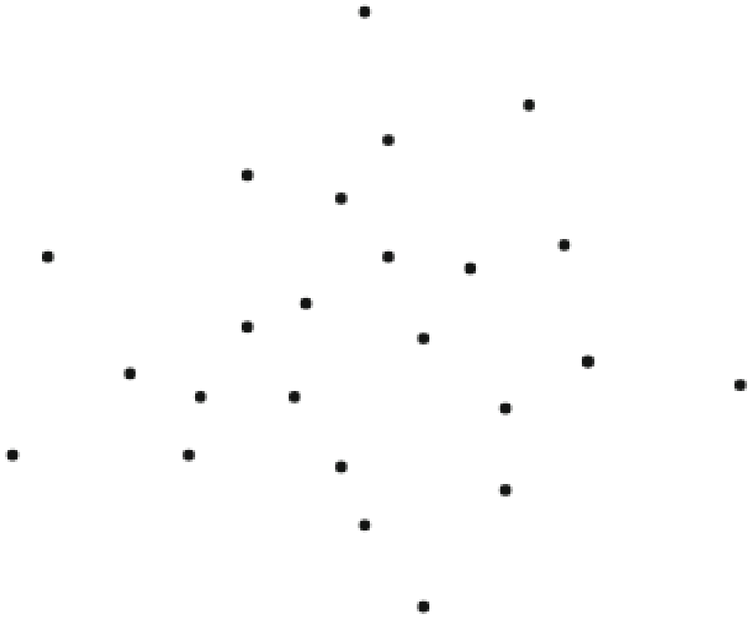
- A depth-based algorithm proceeds in an iterative fashion, where during the *k*-th iteration, all points at the corners of the convex hull of the data set are removed.
- The index of the iteration *k* provides an outlier score
  - Smaller values indicates a greater tendency for a data point to be an outlier.
  - All data points with depth at most *r* can be considered outliers.
  - *r* value can be determined by univariate extreme value analysis

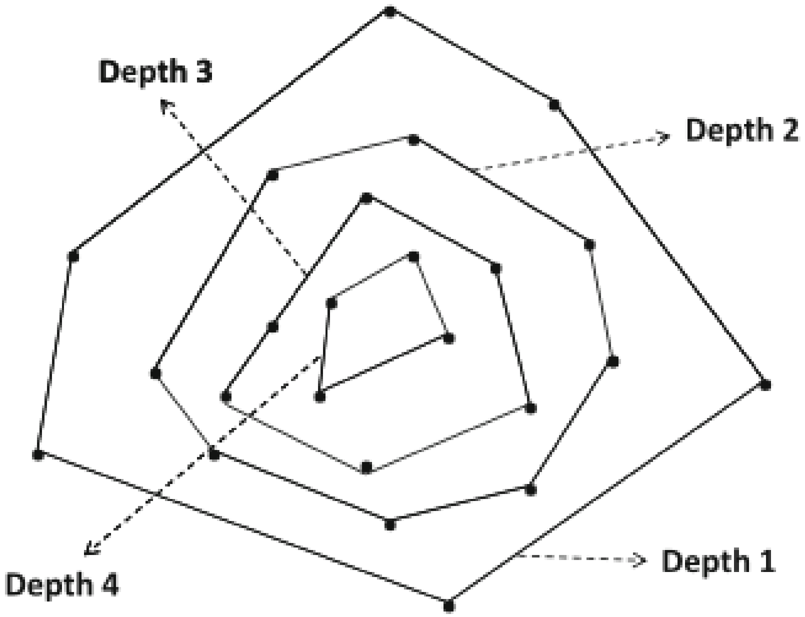
Depth-Based Methods

Algorithm $FindDepthOutliers$(Data Set: $\mathcal{D}$, Score Threshold: r)
begin
  $k = 1$;
  repeat
    Find set S of corners of convex hull of $\mathcal{D}$;
    Assign depth k to points in S;
    $\mathcal{D} = \mathcal{D} - S$;
    $k = k + 1$;
  until(D is empty);
  Report points with depth at most r as outliers;
end

Depth-Based Methods

- The process can be viewed as analogous to peeling the different layers of an onion where the outermost layers define the outliers.





Depth-Based Methods

- This method has the following limitations:
  - No data normalization from statistical perspective
  - All points are treated equally
  - Many data points are indistinguishable
  - The computational complexity increases significantly with dimensionality

Probabilistic Models

- Related to Probabilistic Model-Based Clustering
- Assume data is generated from a mixture-based generative model
- Learn the parameter of the model from data
  - EM algorithm
- Evaluate the probability of each data point being generated by the model
  - Points with low values are outliers

Probabilistic Models

- Data was generated from generative model $\mathcal{M}$ a mixture of k distributions with the probability distributions $\mathcal{G}_1, \ldots \mathcal{G}_k$
  - $\mathcal{G}_i$ is a cluster/mixture component

- Each point in the data set is generated as follows:
  - Select a mixture component with prior probability $\alpha_i = P(\mathcal{G}_i)$ where $i \in \{1 \ldots k\}$.
  - Assume that the rth one is selected
  - Generate a data point from $\mathcal{G}_r$.

Probabilistic Models

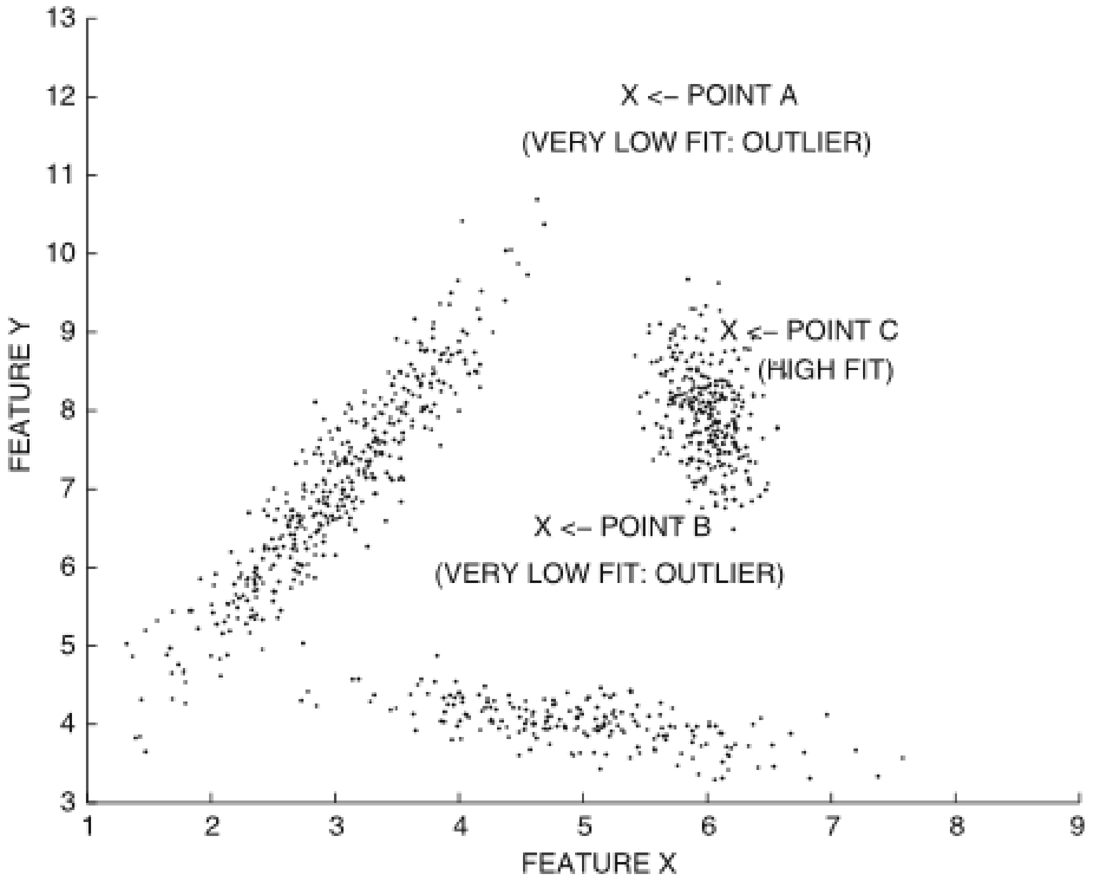
- Parameters of $\mathcal{M}$ generative models are learned via maximum-likelihood criterion to better represents the data points as mixture of distributions
- The probability that $\bar{X_j}$ is generated by $\mathcal{M}$ is given by

$$f^{point}(\overline{X_j}|\mathcal{M}) = \sum_{i=1}^{k} \alpha_i \cdot f^i(\overline{X_j}).$$

- Outliers have a poor probability to have been generated

Probabilistic Models

- Data points A and B will typically have very low fit to the mixture model and will be considered outliers
  - A and B do not naturally belong to any of the mixture components.
- Data point C will have high fit to the mixture model and will, therefore, not be considered an outlier.



Clustering for Outlier Detection

- *Clustering* is about finding *crowds* of data points.
- *Outlier* analysis is about finding data points that are far away from these crowds.

- Clustering and outlier detection share a complementary relationship

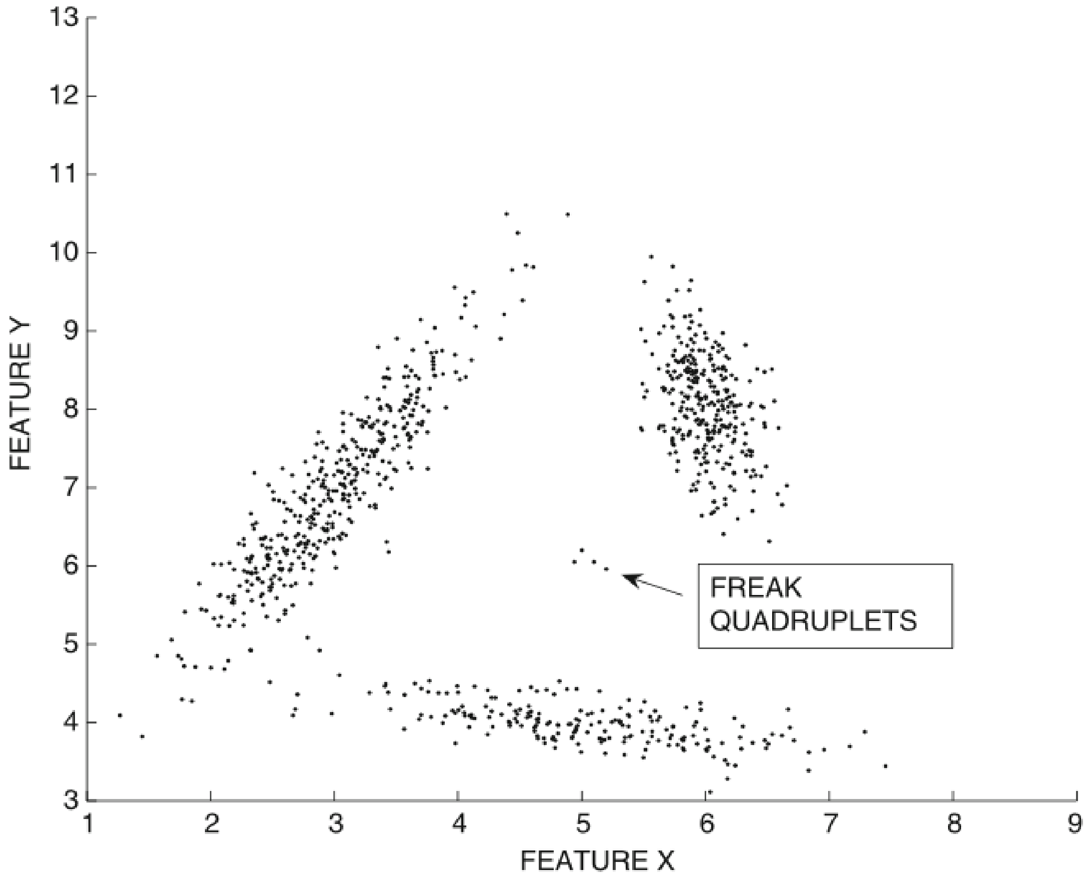
- Every data point can be:
  - A member of a cluster
  - An outlier

Clustering for Outlier Detection

- Can we use a Clustering Algorithm to detect outliers?
  - Clustering algorithms often have an "outlier handling" option that removes data points outside the clusters.
    - Outlier detection is a side-product
    - Such algorithms are not optimized outlier detection.
  - Outliers often tend to occur in small clusters of their own.

Clustering for Outlier Detection

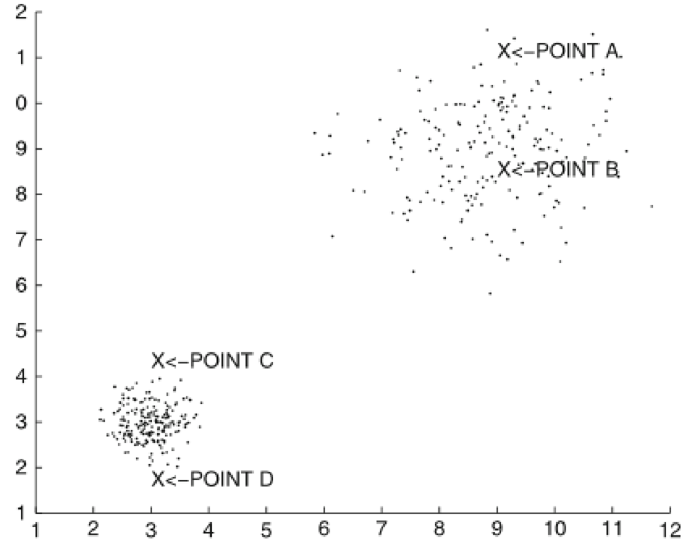
Clustering methods are generally robust to such scenarios because such groups often do not have the critical mass required to form clusters of their own.

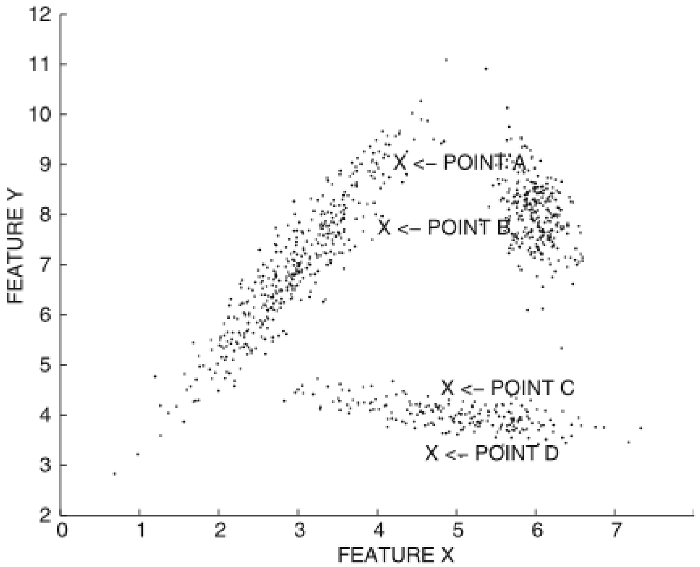


Clustering for Outlier Detection

- A simple method:
  1. Cluster the data
  2. Define the outlier score as the distance of the data point to its cluster centroid

- But it is conditioned on local structure of each cluster
  - Local density
  - Local orientation





Clustering for Outlier Detection

- A better method:
  1. Cluster the data
  2. Define the outlier score as the local Mahalanobis distance

     Suppose $\bar{X}$ belongs to cluster r

$$Maha(\overline{X}, \overline{\mu_r}, \Sigma_r) = \sqrt{(\overline{X} - \overline{\mu_r})\Sigma_r^{-1}(\overline{X} - \overline{\mu_r})^T}$$

$\overline{\mu_r}$ is the mean vector of the r-th cluster

Σ_r is the covariance matrix of the r-th cluster

- This distance is reported as the outlier score.
  - Larger values of the outlier score indicate a greater outlier tendency.
- Once the outlier score has been determined, univariate extreme value analysis can be performed

Clustering for Outlier Detection

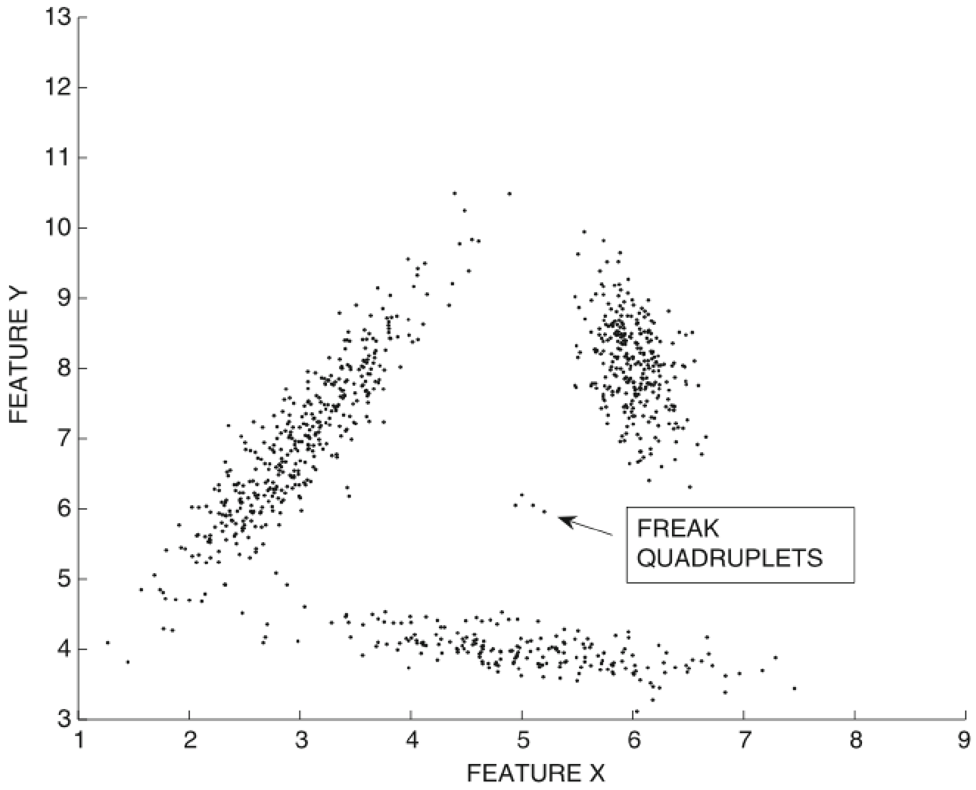
- The use of the local Mahalanobis distance has an interesting connection to the likelihood fit criterion of EM algorithm
  - EM algorithm: Mahalanobis distance of $\bar{X}$ to different mixture component means (cluster means)
  - Outlier score based on this local distance
- Clustering methods are based on global analysis:
  - Small, closely related groups of data points will not form their own clusters in most cases.

Clustering for Outlier Detection

- Most clustering algorithms require a minimum critical mass for a set of data points to be considered a cluster → high outlier score.
- Clustering methods can detect small and closely related groups of data points → outliers.
  - Not true for some density-based methods based purely on local analysis.
- Clustering algorithms falls in distinguish between
  - a data point that is ambient noise
  - a data point that is a truly isolated anomaly

Distance-Based Outlier Detection

- The distance-based outlier score of an object O is its distance to its *k*th nearest neighbor.
  - The value of *k* is a user-defined parameter
  - *k* > 1 helps identify isolated groups of outliers.
  - target data point, for which the outlier score is computed, is itself not included among its *k*-nearest neighbors.



k > 3 for outlier detection

Distance-Based Outlier Detection

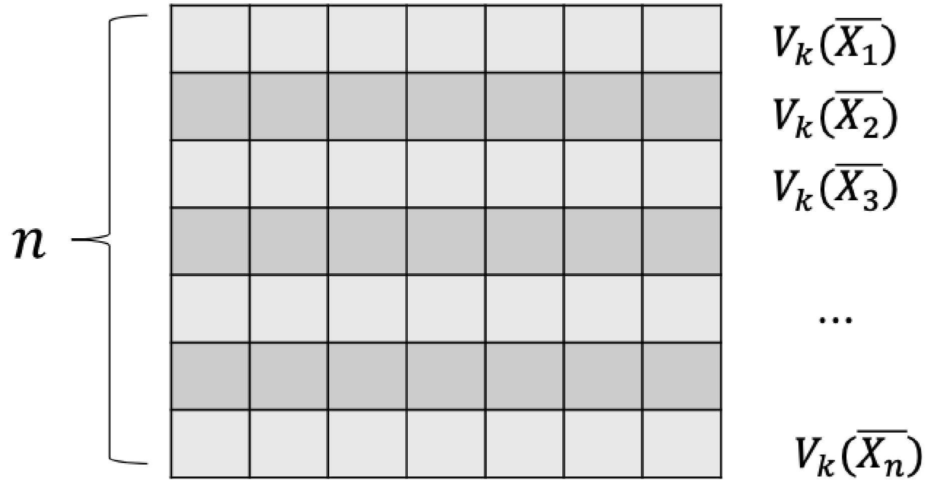
- Distance-based methods typically use a finer granularity of analysis than clustering methods and can therefore distinguish between *ambient noise* and *truly isolated anomalies*.
  - Ambient noise will typically have a lower k-nearest neighbor distance than a truly isolated anomaly.
- For better granularity, higher computational complexity.
  - Index structure → effective when the dimensionality is low
  - Pruning tricks → when only the top-*r* outliers are needed
  - Those techniques can be combined

Pruning Methods

- Pruning methods are used only for the case where the top-r ranked outliers need to be returned, and the outlier scores of the remaining data points are irrelevant.
- Those methods can be used only for the binary-decision version of the algorithm.
- The basic idea in pruning methods is to reduce the time required for the *k*-nearest neighbor distance computations by ruling out data points quickly that are obviously nonoutliers even with approximate computation.

Pruning Methods

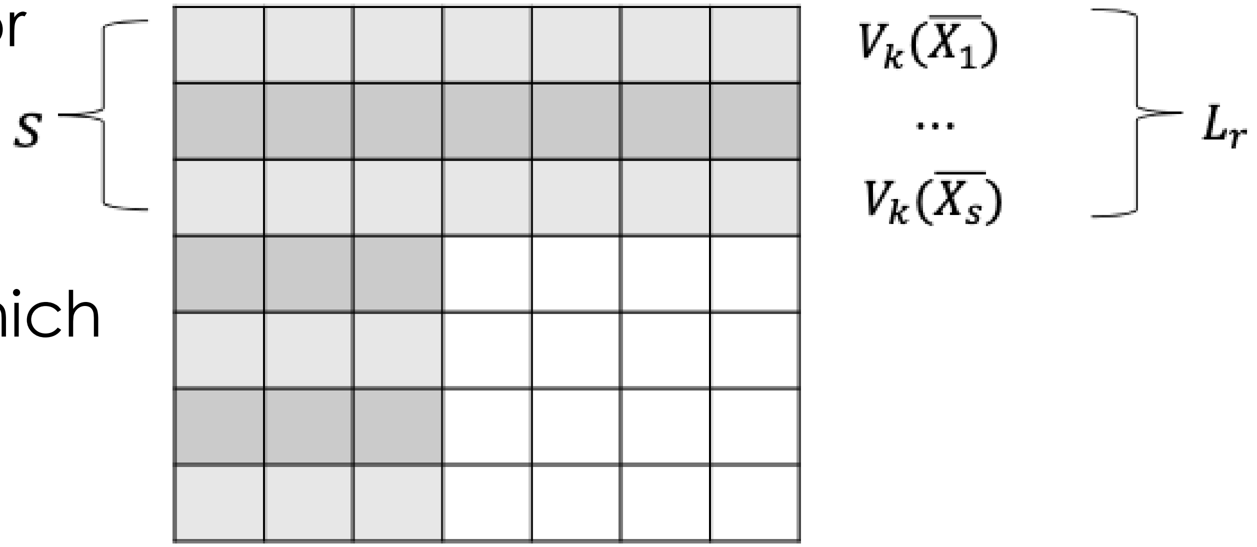
- Naïve approach for determining top *r*-outliers
  1. Evaluate the $n \times n$ distance matrix



  2. Find the k-th smallest value in each row
  3. Choose r data points with largest $V_k(\cdot)$

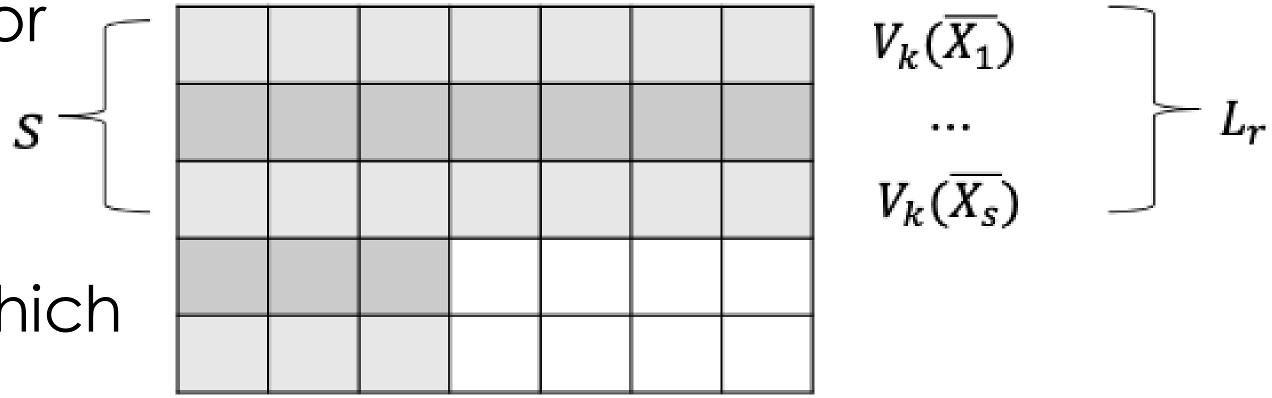
Pruning Methods

- Sampling
  1. Evaluate a $s \times n$ distance matrix where $s \ll n$
  2. Find the k-th smallest value in each row
  3. Identify the r-th score in top s-rows, *local outliers*
     L_r is a *lower bound* for outliers, points with lower values are *for sure* not outliers
  4. Remove points for which estimated distance is lower than L_r



Pruning Methods

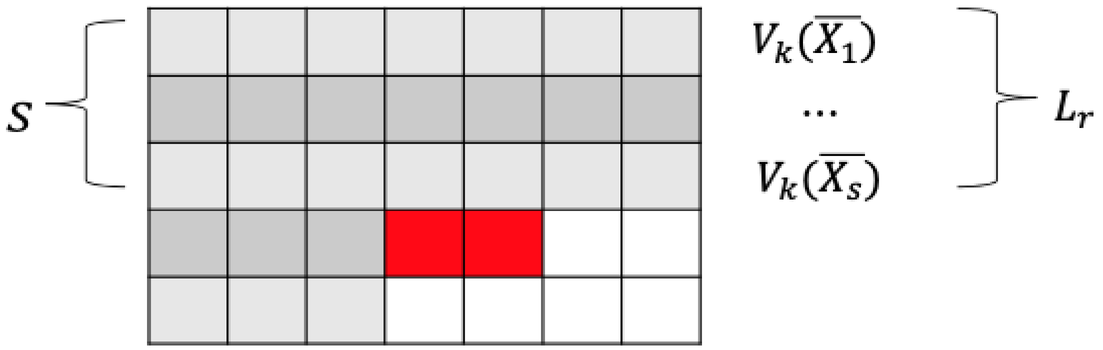
- Sampling
  1. Evaluate a $s \times n$ distance matrix where $s \ll n$
  2. Find the k-th smallest value in each row
  3. Identify the r-th score in top s-rows, *local outliers*
     L_r is a *lower bound* for outliers, points with lower values are *for sure* not outliers
  4. Remove points for which $\widehat{V_k}(\cdot) \leq L_r$
  5. Calculate missing distances to find real top-r outliers



Pruning Methods

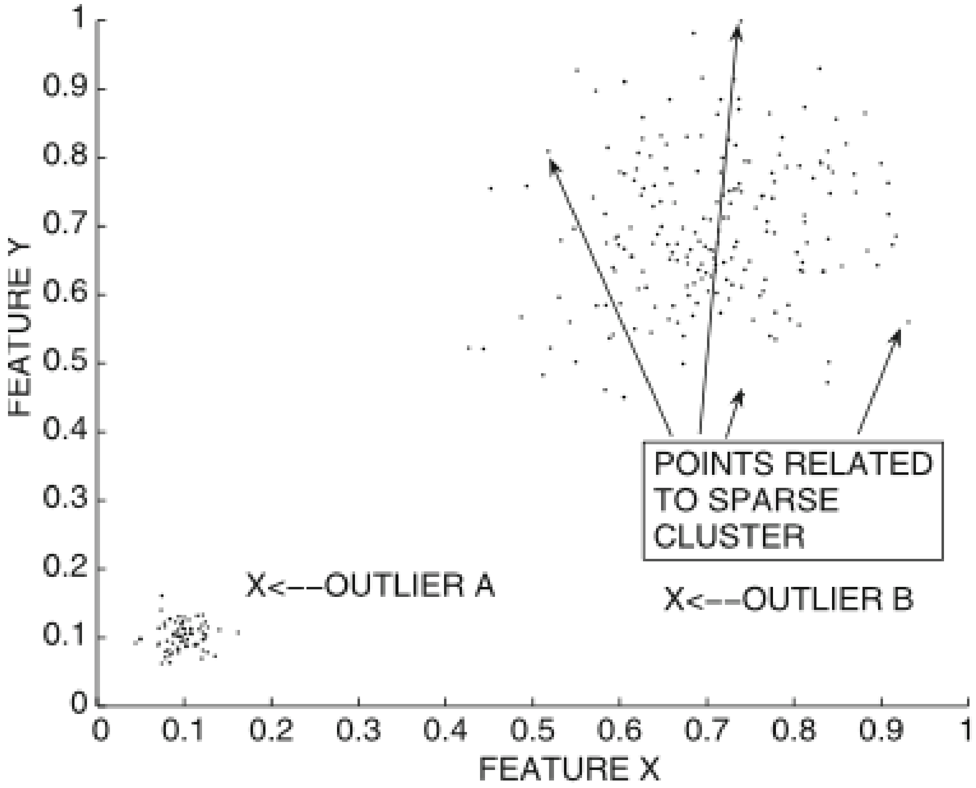
- Early Termination Trick with Nested Loops
    5. Calculate missing distances to find real top-r outliers *but STOP* if $\widehat{V_k}(\cdot) \leq L_r$ or update L_r if needed

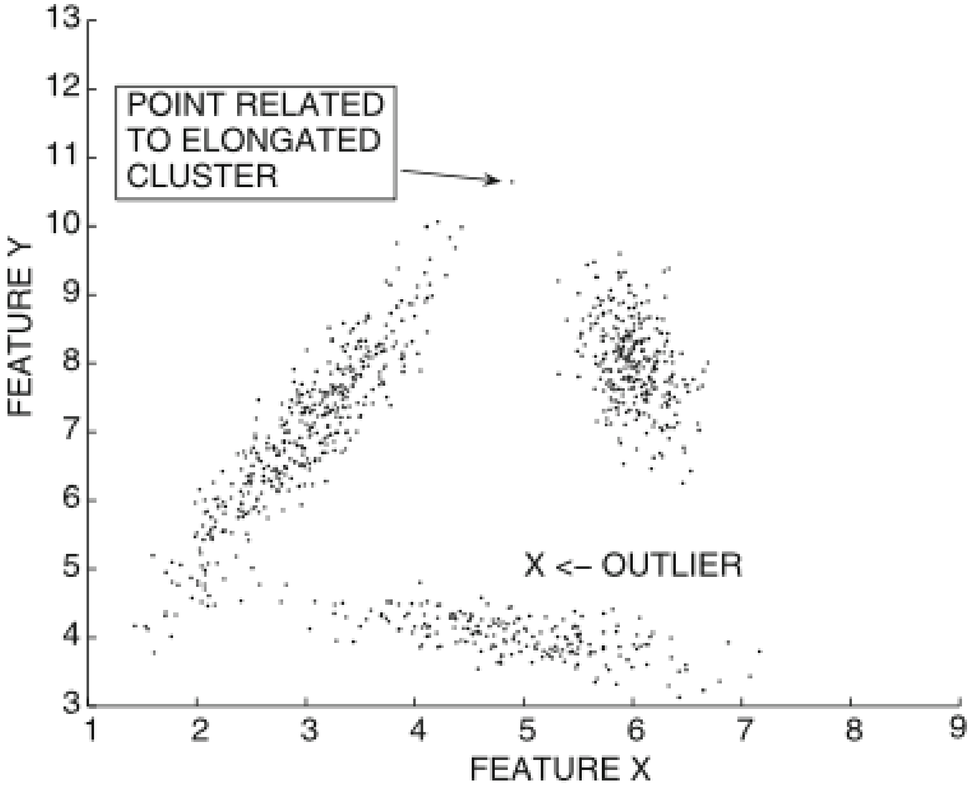
for each $\overline{X} \in \mathcal{R}$ **do begin**
  for each $\overline{Y} \in \mathcal{D} - \mathcal{S}$ **do begin**
    Update current k-nearest neighbor distance estimate $V^k(\overline{X})$ by
      computing distance of $\overline{Y}$ to $\overline{X}$;
    if $V^k(\overline{X}) \leq L$ **then** terminate inner loop;
  endfor
  if $V^k(\overline{X}) > L$ **then**
    include $\overline{X}$ in current r best outliers and update L to
      the new rth best outlier score;
endfor



Local Distance Correction Methods

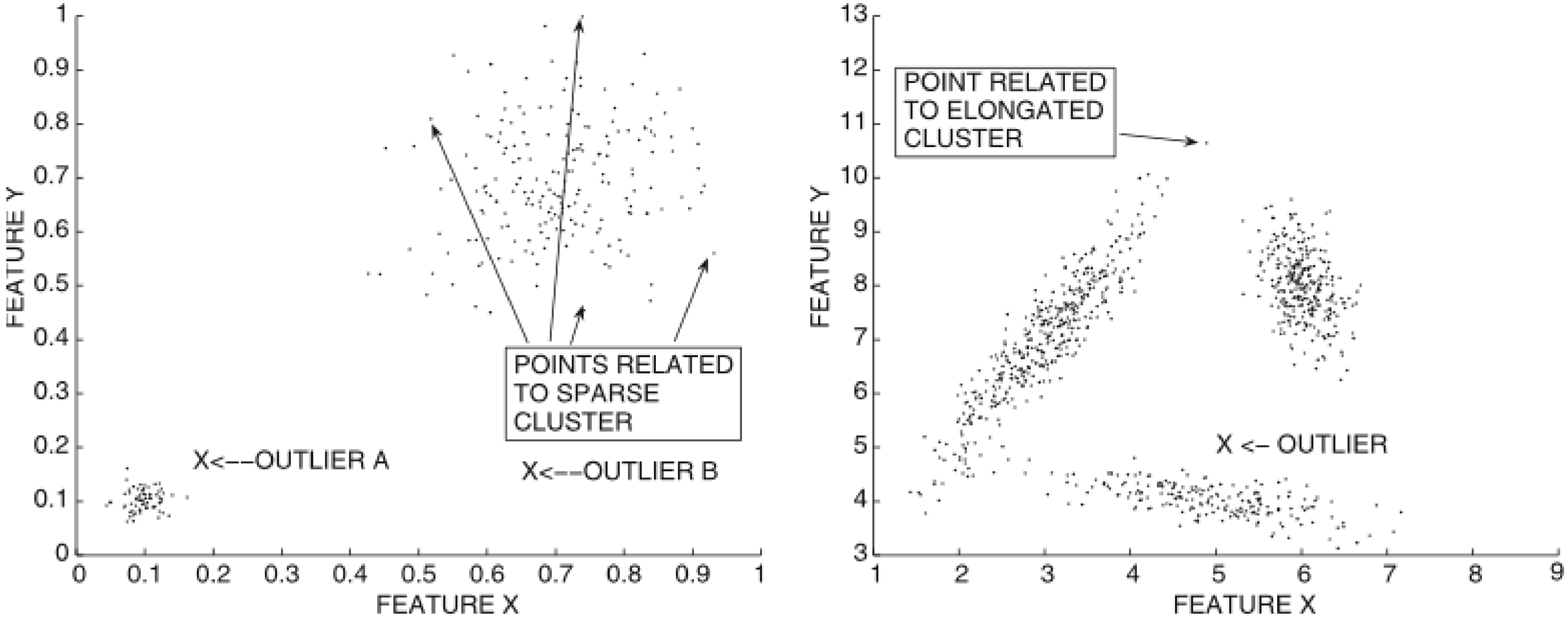
- Euclidean distance, like gaussian, what else?





Local Distance Correction Methods

- Euclidean distance, like gaussian, what else? Spoiler, NO

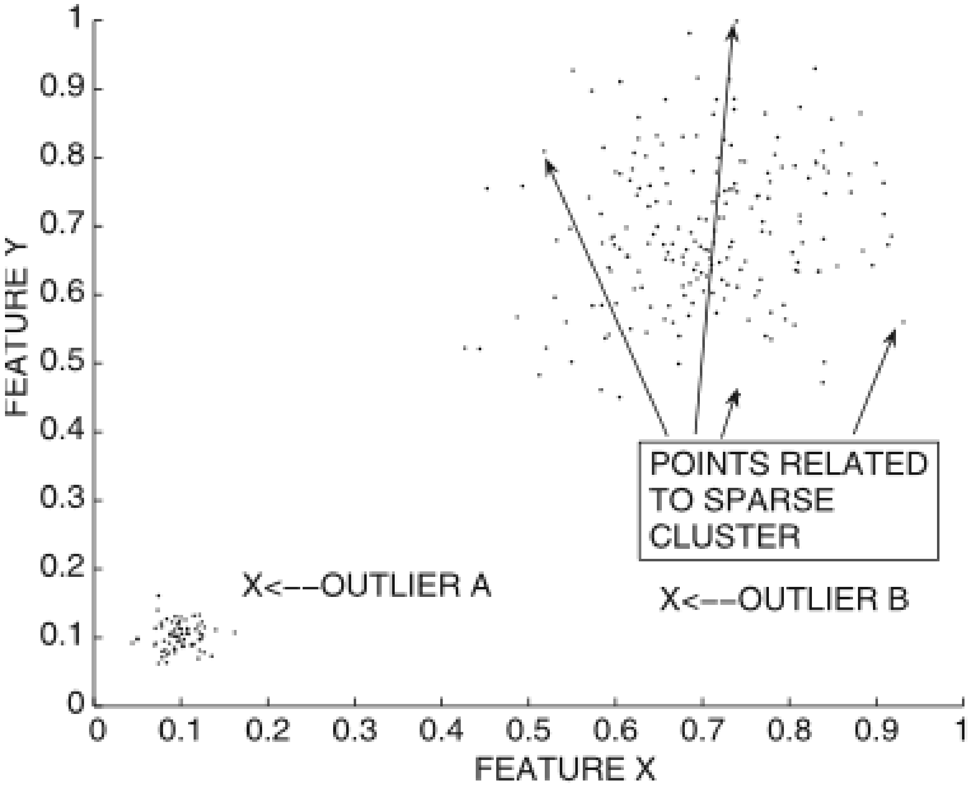


Local Distance Correction Methods

- Straightforward measures do not reflect the intrinsic distances between data points when the density and shape of the clusters vary significantly with data locality.
- A normalization step may be required for leverage the local structure of clusters
- Two approaches
  - Local Outlier Factor (LOF)
  - Instance-Specific Mahalanobis Distance

Local Distance Correction Methods

- Local Outlier Factor (LOF)
  - It adjusts for local variations in cluster density by normalizing distances with the average point-specific distances in a data locality.
  - It is a normalized distance-based approach where the normalization factor corresponds to the average local data density.



Local Distance Correction Methods

- Local Outlier Factor (LOF)
  - Let $V^k(\bar{X})$ be the distance of $\bar{X}$ to its k-nearest neighbor
  - Let $L_k(\bar{X})$ be the set of points within the k-nearest neighbor distance of $\bar{X}$
  - Reachability Distance

$$R_k(\bar{X}, \bar{Y}) = \max\{Dist(\bar{X}, \bar{Y}), V^k(\bar{X})\}$$

    - It is not symmetric between $\bar{X}$ and $\bar{Y}$
    - If $Dist(\bar{X}, \bar{Y})$ is large, $R_k(\bar{X}, \bar{Y}) = Dist(\bar{X}, \bar{Y})$
    - Otherwise $R_k(\bar{X}, \bar{Y}) = V^k(\bar{X})$, where $V^k(\bar{X})$ smooths the method and provides stability

Local Distance Correction Methods

- Local Outlier Factor (LOF)
  - Average Reachability Distance, $L_k(\bar{X})$ is $\bar{X}$ neighborhood of size k

$$AR_k(\bar{X}\ ) = \mathrm{MEAN}_{\bar{Y}\in L_k(\bar{X})} R_k(\bar{X},\bar{Y})$$

  - Local Outlier Factor

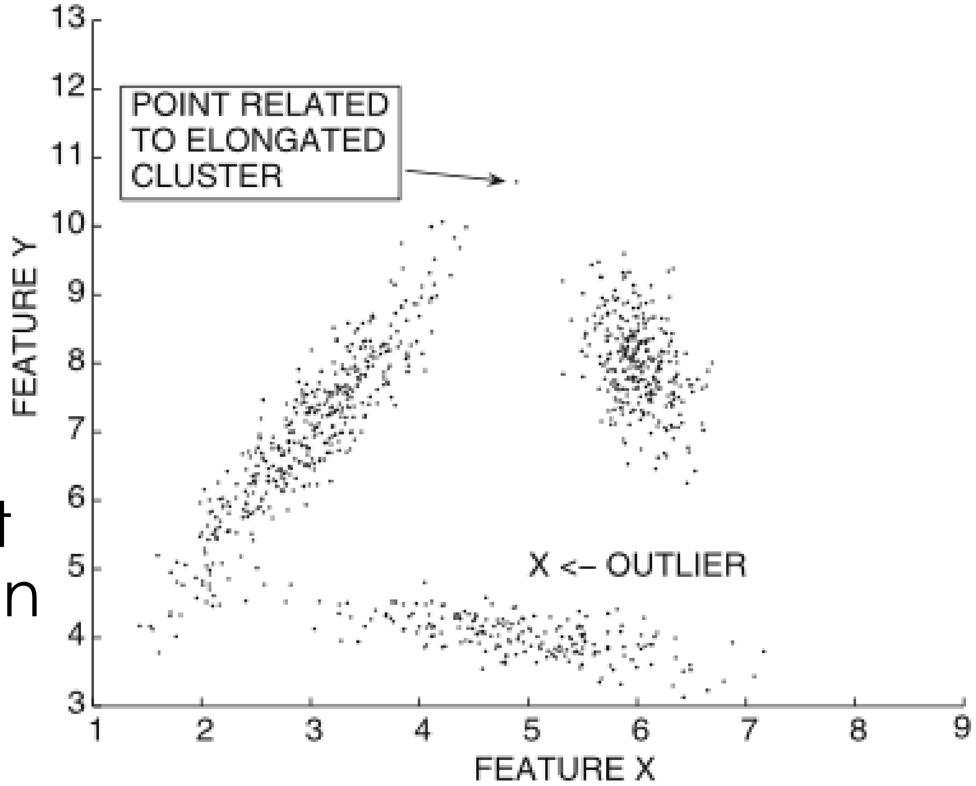
$$LOF_k(\bar{X}) = \mathrm{MEAN}_{\bar{Y}\in L_k(\bar{X})} \frac{AR_k(\bar{X})}{AR_k(\bar{Y})}$$

    - LOF value is larger for outliers
    - LOF value close to 1 for other points
  - The outlier score is calculated over a range of k values

$$\max_k LOF_k(\bar{X})$$

Local Distance Correction Methods

- Instance-Specific Mahalanobis Distance
  - It is designed for adapt to varying local shapes.
  - The idea is to define a local Mahalanobis distance for each data point: it will be based on its covariance matrix and relative shape.
  - The neighborhood of a data point is hard to define with the Euclidean distance for no-spherical neighborhood.



Local Distance Correction Methods

- Instance-Specific Mahalanobis Distance
  - An agglomerative approach is used to determine the k-neighborhood $L_k(\bar{X})$ of $\bar{X}$:
  - $\bar{X}$ is added to $L_k(\bar{X})$.
  - Data points are iteratively added to $L_k(\bar{X})$ have the smallest distance to their closest point in $L_k(\bar{X})$.
  - Such an approach tends to *grow* the neighborhood with the same shape as the cluster.

Local Distance Correction Methods

- Instance-Specific Mahalanobis Distance
  - Instance-specific Mahalanobis score is the Mahalanobis distance of $\bar{X}$ to the mean $\overline{\mu_k(X)}$ of data points in $L_k(\bar{X})$.

$$LMaha_k(\bar{X}) = Maha(\bar{X}, \overline{\mu_k(X)}, \Sigma_k(\bar{X}))$$

  - The outlier score is calculated over a range of *k* values

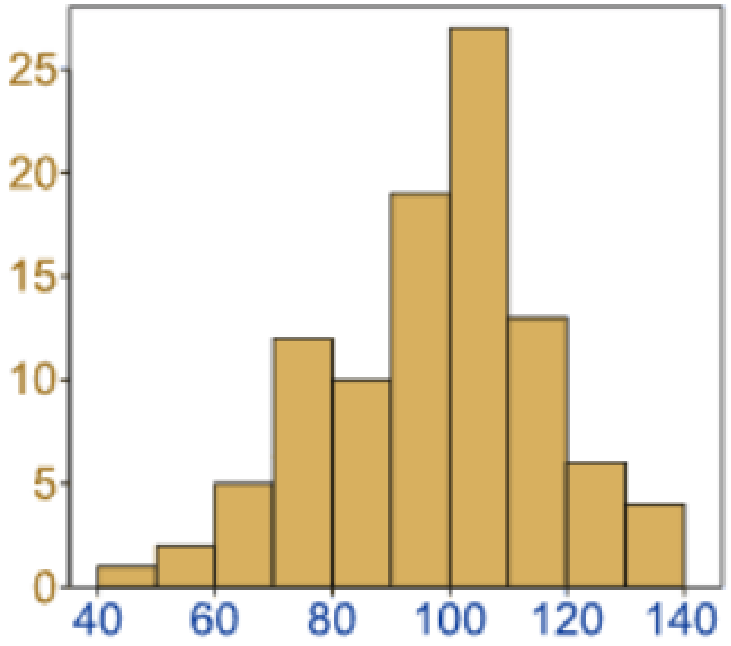
$$\max_k LMaha_k(\bar{X})$$

Density-Based Methods

- Density-based methods are based on similar principles as density-based clustering.
- Strategies
  - Histogram-based techniques (1d grid)
  - Grid-Based techniques
  - Kernel Density Estimation
- These methods did not found popularity because of their difficulty in adjusting to local density variations.
- The definition of density becomes more challenging with increasing dimensionality.
- More frequently used in the univariate case due to their natural probabilistic interpretation.

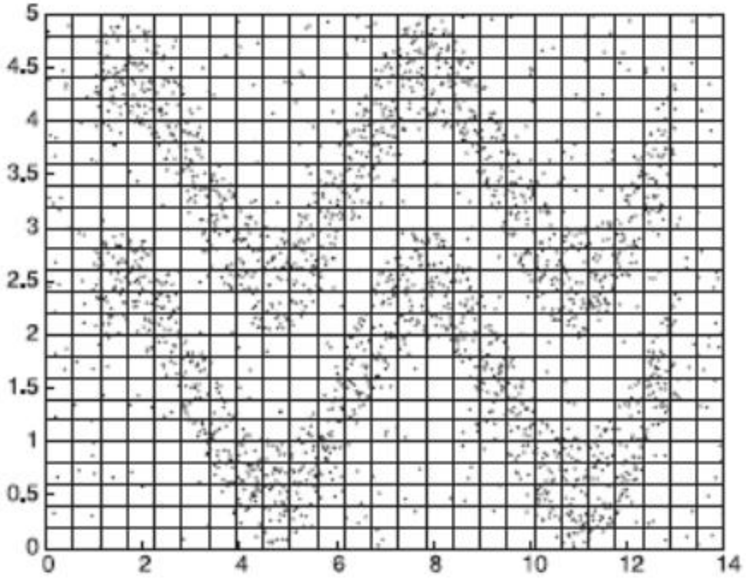
Histogram-based techniques

- Histogram for 1-dimensional data
  - Data is discretized into bins
  - Frequency of each bin is estimated.
  - Data points that lie in bins with very low frequency are reported as outliers.
  - The count for a bin does not include the point itself to minimize overfitting for smaller bin widths or smaller number of data points.



Grid-based techniques

- Grid for high-dimensional data
  - Generalization with grid-structure
  - Each dimension is partitioned into *p* equi-width ranges.
  - The number of points in a particular grid region is reported as the outlier score.
  - Data points that have density less than τ in any grid region are reported as outliers.
  - The appropriate value of τ can be determined by using univariate extreme value analysis.



Histogram- and Grid-based techniques

- Challenges
  - Size of histograms and grids
  - Histograms are too local and do not consider a global view
  - Sparsity
    - Density of the grid only depends on the data points inside it, and an isolated group of points may create an artificially dense grid cell with high granularity of representation
    - If the density distribution varies significantly with data locality, grid-based methods may find it difficult to normalize for local variations in density
  - Those methods do not work very well in high dimensionality

Kernel Density Estimation

- In Kernel Density Estimation (KDE), a continuous estimate of the density is generated at a given point.
- Density value at a given point is estimated as the sum of the smoothed values of kernel functions $K_h(\cdot)$ associated with each point in the data set.
- Each kernel function is associated with a kernel width h that determines the level of smoothing created by the function.

Kernel Density Estimation

- The kernel estimation $f(\bar{X})$ based on n data points of dimensionality d, and kernel function $K_h(\cdot)$ is

$$f(\bar{X}) = \frac{1}{n} \cdot \sum_{i=1}^{n} K_h(\bar{X} - \bar{X}_i)$$

- A gaussian kernel is used

$$K_h(\bar{X} - \bar{X}_i) = \left(\frac{1}{\sqrt{2\pi} \cdot h}\right)^d e^{-\frac{||\bar{X} - \bar{X}_i||^2}{2h^2}}$$

Kernel Density Estimation

- The estimation error is defined by the kernel width *h*, chosen in a data-driven manner.
- For most smooth functions, when the number of data points goes to infinity, the estimate asymptotically converges to the true density value, if *h* is chosen appropriately.
- The density is computed without including the point itself in the density computation.
- Obtained density values is the outlier score.
  - Low values indicate greater tendency to be an outlier.
- Challenging for wide local density variations

Information-Theoretic Models

How can be represented?

ABABABABABABABABABABABABABABABABAB (1)

ABABACABABABABABABABABABABABABABAB (2)

Information-Theoretic Models

How can be represented?

ABABABABABABABABABABABABABABABABAB (1) → *AB17*

ABABACABABABABABABABABABABABABAB (2) → *AB2A1C1AB14*

In the second case there is a substantial increase in its minimum description length due to an outlier.

Information-Theoretic Models

- Information-theoretic models are based on this general principle because they measure the increase in model size required to describe the data as concisely as possible.

- Conventional Methods
  - Fix model, then calculate the deviation
  - Outliers are defined on the basis of a "summary" model of normal patterns. When a data point deviates significantly from the estimations of the summary model, then this deviation value is reported as the outlier score.
  - Clearly, a trade-off exists between the size of the summary model and the level of deviation.
  - e.g. for a clustering model, a larger number of cluster centroids (model size) will result in lowering the maximum deviation of any data point (including the outlier) from its nearest centroid.

Information-Theoretic Models

- Information-Theoretic Models
  - Fix the deviation, then learn the model
  - Outlier score of $\bar{X}$: increase of the model size when $\bar{X}$ is present
    - Fix the maximum allowed deviation (instead of the number of cluster centroids), compute the number of cluster centroids required to achieve the same level of deviation, with and without a particular data point.
    - This increase is reported as the outlier score in the information-theoretic version of the same model.

Information-Theoretic Models

- Information-Theoretic Models
  - The idea here is that each point can be estimated by its closest cluster centroid, and the cluster centroids serve as a "code-book" in terms of which the data is compressed in a lossy way.
  - Information-theoretic models can be viewed as a complementary version of conventional models in which a different aspect of the space-deviation trade-off curve is examined.
  - Every conventional model can be converted into an information-theoretic version by examining the space-deviation.

Information-Theoretic Models

- Probabilistc models

Conventional Method	Information-Theoretic Method
• Learn the parameters of generative model with a fixed size • Use the fit of each data point as the outlier score	• Fix a maximum allowed deviation (a minimum value of fit) • Learn the size and values of parameters • Increase of size is used as the outlier score

- Clustering or density-based models

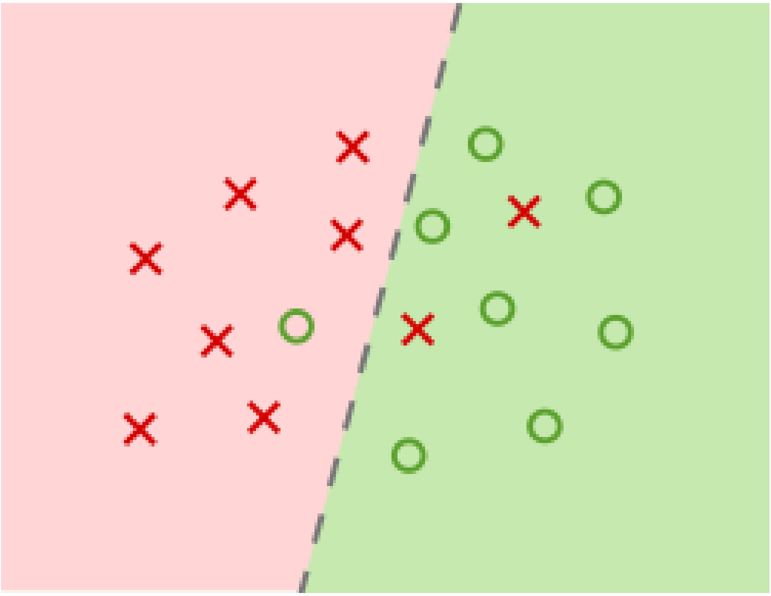
Conventional Method	Information-Theoretic Method
• Number of bins/clusters is fixed (model size) • The error in approximating the data point with a central element of the cluster is the deviation.	• Fix a maximum allowed deviation • Model size is reported as outlier score

Outlier Validity

- Outlier Validity for evaluating obtained outliers
  - It is much harder in outlier detection than data clustering.
- Methodological Challenges
  - Unsupervised problem and internal criteria are rarely used
  - A particular validity measure will favor an algorithm using a similar objective function criterion
    - overfitting issues may be overcome leveraging a validity measure different from the outlier detection models used.
- External Measures
  - The known outlier labels from a synthetic data set
  - The rare class labels from a real data set
  - A *threshold* is typically used on the outlier score to generate a binary label.

Outlier Validity

- Used to determine a trade-off curve between false-positive and false negatives, as determined by the chosen threshold
- The red crosses within the green area represent false positives, negative samples that were classified as positive.
- The green circles within the red area represent false negatives, positive samples that were classified as negative.



Outlier Validity

- Given a threshold t, denote the set of outliers by $\mathcal{S}(t)$
- $\mathcal{G}$ is the set of outliers (ground-truth)
- True-positive rate (recall)
  - Percentage of ground-truth outliers that have been reported as outliers at threshold t

$$TPR(t) = Recall(t) = 100 * \frac{|\mathcal{S}(t) \cap \mathcal{G}|}{|\mathcal{G}|}$$

- False-positive rate
  - Percentage of the falsely reported positives out of the ground-truth negatives.

$$FPR(t) = 100 * \frac{|\mathcal{S}(t) - \mathcal{G}|}{|\mathcal{D} - \mathcal{G}|}$$

Outlier Validity

- Receiver Operating Characteristic (ROC) curve
  - Plot $TPR(t)$ versus $FPR(t)$
  - Curve that strictly dominates another, is associated to superior algorithm
    Not use ROC curve for tuning the parameters of the algorithm

$|\mathcal{D}| = 100$
$|\mathcal{G}| = 5$
$|\mathcal{D} - \mathcal{G}| = 95$

Algorithm	Rank of ground-truth outliers
Algorithm A	1, 5, 8, 15, 20
Algorithm B	3, 7, 11, 13, 15
Random Algorithm	17, 36, 45, 59, 66
Perfect Oracle	1, 2, 3, 4, 5